



Project title:

Predicting Food Delivery Times Using Machine Learning

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ARTIFICIAL INTELLIGENCE

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Introduction

This project focuses on predicting food delivery times based on various influencing factors such as distance, weather conditions, traffic levels, delivery type, and courier experience. The aim is to classify whether an order will arrive on time or late using multiple machine learning approaches. Models applied include Logistic Regression, Decision Tree, Naive Bayes, and a Neural Network, along with KMeans clustering for unsupervised learning. The motivation is to improve delivery efficiency, optimize logistics planning, and enhance customer satisfaction by identifying potential delays in advance.

Dataset description

Source: [Food delivery times](#)

Dataset details:

The dataset contains 8 input features:

- Distance_km (quantitative)
- Weather (categorical)
- Traffic_Level (categorical)
- Time_of_Day (categorical)
- Vehicle_Type (categorical)
- Preparation_Time_min (quantitative)
- Courier_Experience_yrs (quantitative)
- Order_ID (identifier, not useful for prediction)

The target feature is Delivery_Speed (categorical: Fast, Average, Slow).

Total Data Points: 1000 rows

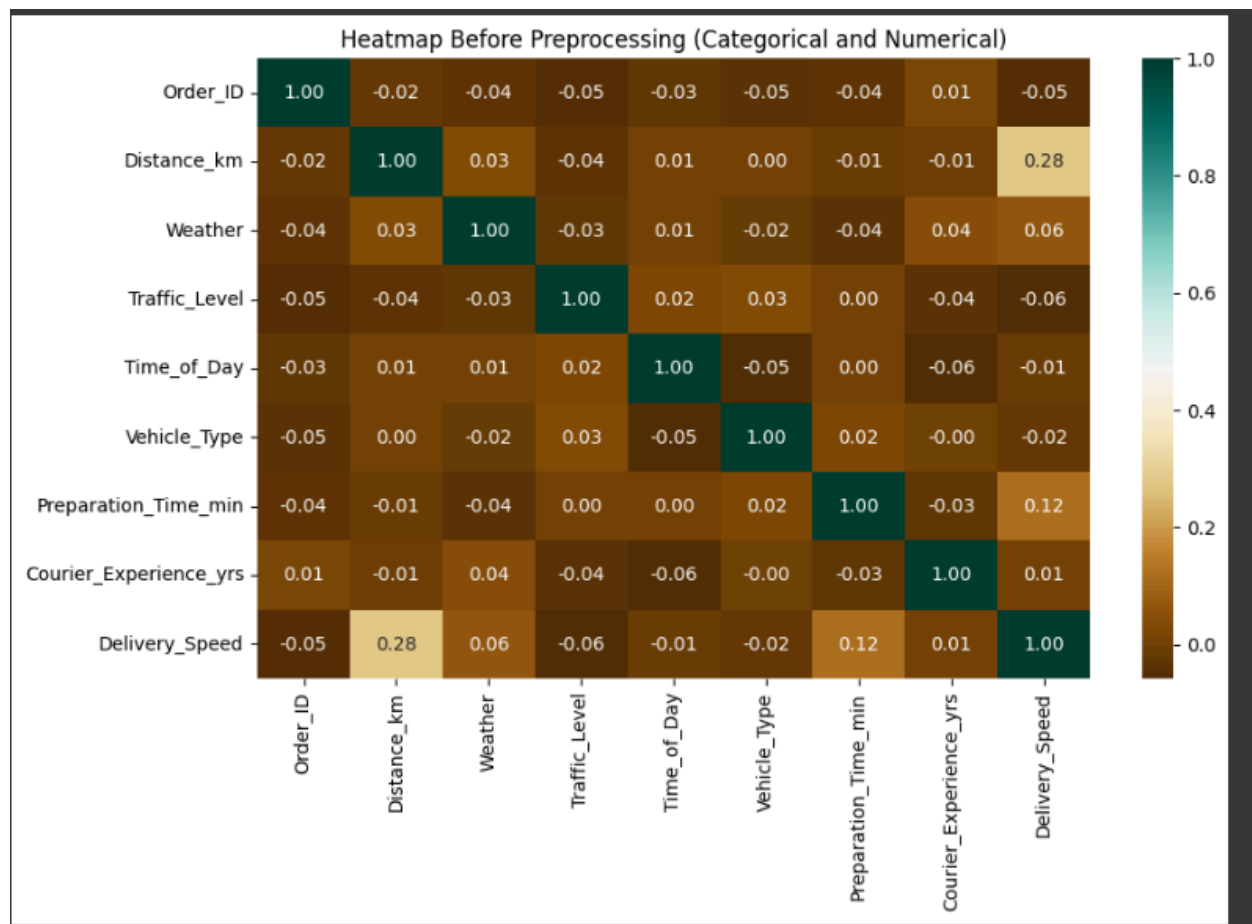
This is a **classification** problem because the output variable (Delivery_Speed) is **categorical**.

Encoding Needed for categorical variables like Weather and Traffic_Level(One hot encoding). So they can be processed by machine learning algorithms.

Correlation between features :

The correlation heatmap shows that the strongest relationship is between Distance_km and Delivery_Speed (0.28), indicating that longer distances are more likely to result in slower deliveries. Preparation_Time_min also has a slight positive correlation (0.12) with delivery speed, meaning higher preparation times tend to delay orders. Courier_Experience_yrs shows a weak negative correlation (-0.06), suggesting that more experienced couriers usually deliver faster.

Other features, such as Weather, Traffic_Level, and Time_of_Day, have very low correlations with delivery speed, and correlations among independent features are minimal. This indicates that multicollinearity is not an issue, which is beneficial for predictive modeling.



Imbalanced Dataset

The bar chart reveals that the dataset is not perfectly balanced in terms of the target feature Delivery_Speed. The majority of deliveries fall under the “Average” category, while “Fast” and “Slow” deliveries occur less frequently. This imbalance suggests that models may become biased toward predicting “Average” delivery unless stratified sampling or balancing techniques are applied.



Dataset pre-processing

1. Handling Null/ Missing Values: The dataset had some missing values in both numerical and categorical features.

For numerical columns (Distance_km, Preparation_Time_min, Courier_Experience_yrs), missing values were replaced with the **median** to avoid distortion from outliers.

For categorical columns (Weather, Traffic_Level, Time_of_Day, Vehicle_Type), missing values were replaced with the most frequent category (**mode**) to preserve the overall distribution.

2. **Encoding Categorical Variables:** Since machine learning algorithms require numerical inputs, categorical features were transformed using One-Hot Encoding.

Example: For Weather, categories such as Clear, Rainy, Foggy, Windy were converted into separate binary columns (Weather_Clear, Weather_Rainy, etc.).

Feature Scaling: To ensure that all features contribute equally to model training, scaling was applied:

StandardScaler: Used for most models (Logistic Regression, Neural Networks), transforming features to have mean = 0 and standard deviation = 1.

MinMaxScaler: Applied for Naive Bayes, ensuring values lie in the [0,1] range as required for probability-based calculations.

Order_ID column was dropped as it was not useful for prediction .

Dataset Splitting

The dataset was split into training and testing sets in an 80:20 ratio. The split was stratified, ensuring a balanced distribution of the target classes across both sets.

Model training & testing

The following machine learning models were implemented using sklearn:

1. Logistic Regression – used as a baseline classifier with class balancing.
2. Naive Bayes – applied after MinMax scaling to handle categorical-style data.
3. Neural Network – implemented with TensorFlow/Keras for higher accuracy.
4. KMeans Clustering – applied as an unsupervised method to discover groupings in the dataset.

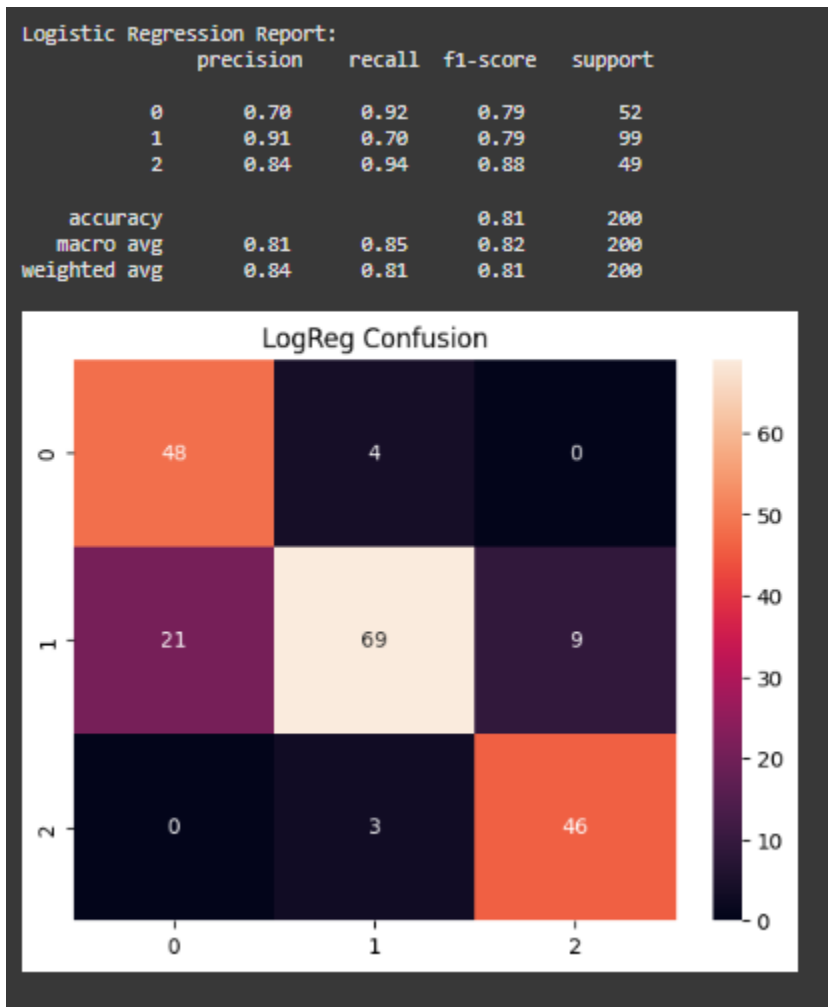
Model selection/Comparison analysis

The comparison of Logistic Regression, Naive Bayes, and Neural Network models for predicting delivery speed shows that Logistic Regression achieved the best overall performance. It had the highest accuracy (81.5%) and the top AUC score (0.94), making

it the most reliable model for this dataset. The Neural Network followed closely with an accuracy of 80.5% and an AUC of 0.93, showing strong predictive power but requiring more computational resources. Naive Bayes performed the weakest, with an accuracy of 55.0% and an AUC of 0.72, indicating difficulties in distinguishing between delivery speed classes.

Confusion Matrix:

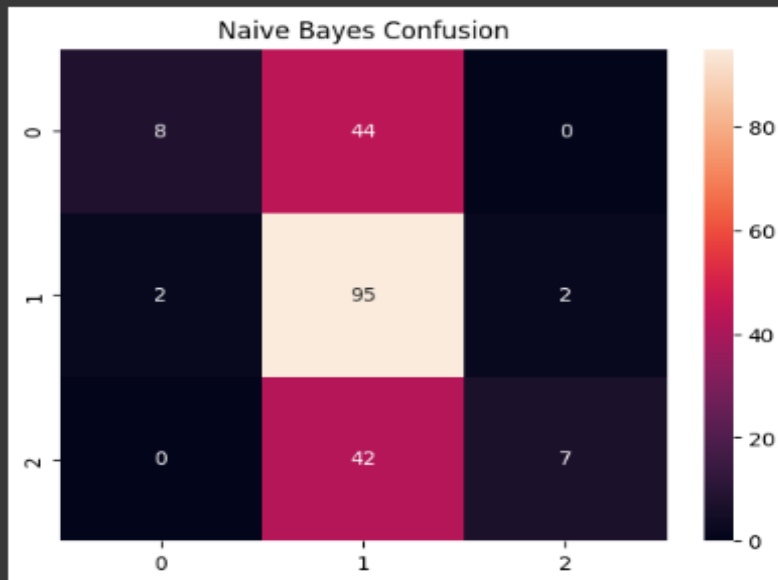
Logistic Regression



Naive Bayes

Naive Bayes Report:

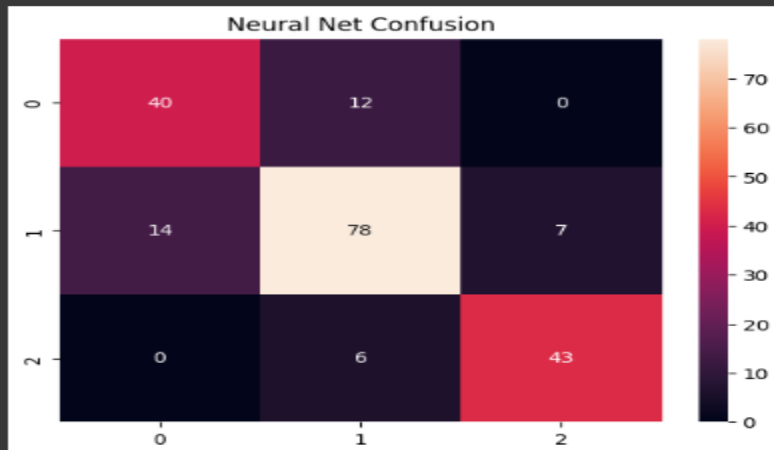
	precision	recall	f1-score	support
0	0.80	0.15	0.26	52
1	0.52	0.96	0.68	99
2	0.78	0.14	0.24	49
accuracy			0.55	200
macro avg	0.70	0.42	0.39	200
weighted avg	0.66	0.55	0.46	200



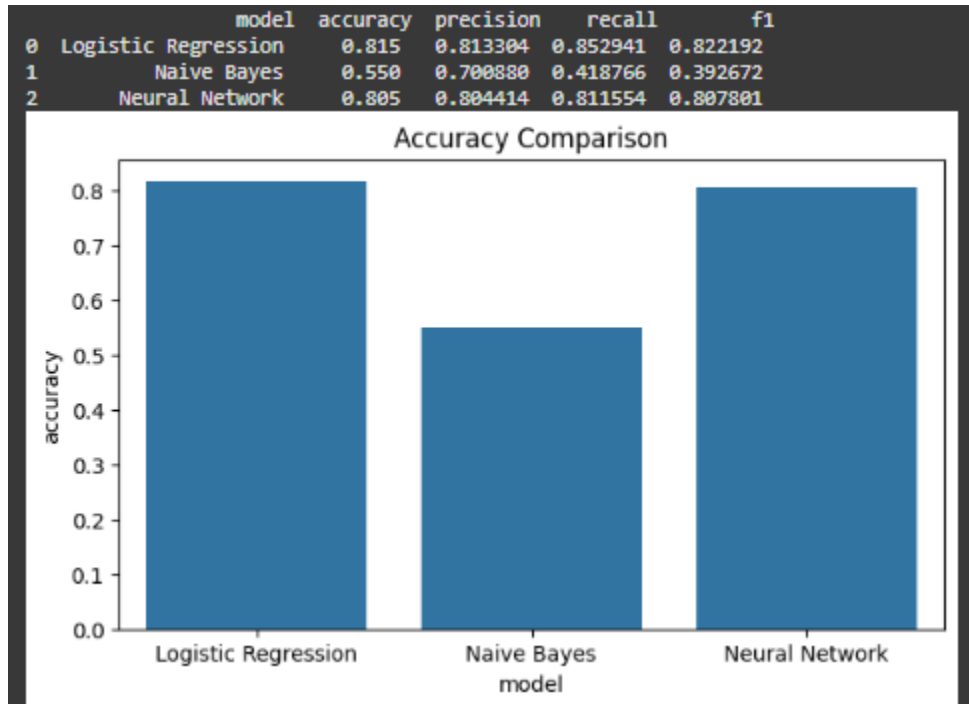
Neural Network

Neural Network Report:

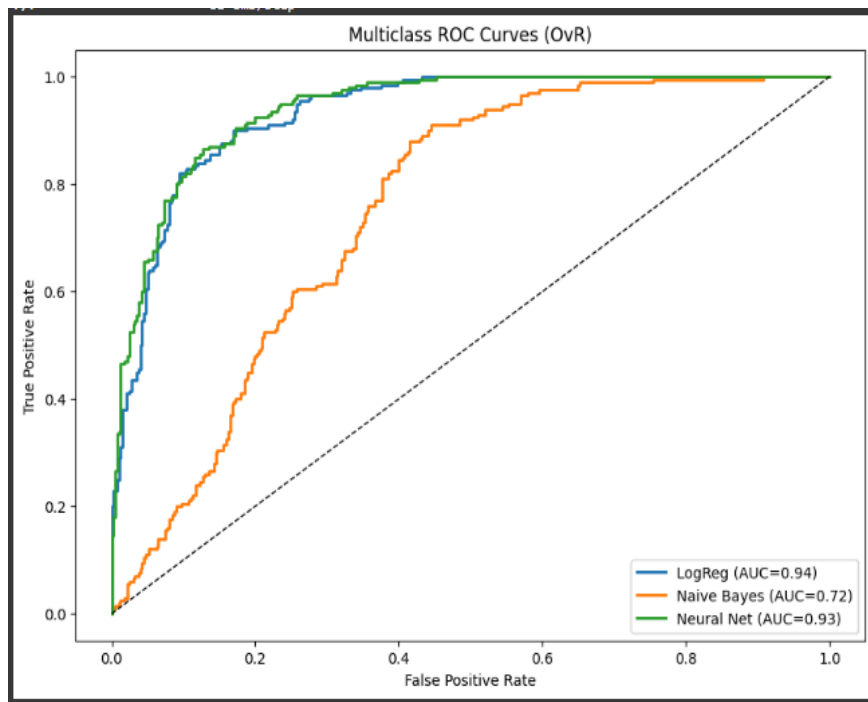
	precision	recall	f1-score	support
0	0.74	0.77	0.75	52
1	0.81	0.79	0.80	99
2	0.86	0.88	0.87	49
accuracy			0.81	200
macro avg	0.80	0.81	0.81	200
weighted avg	0.81	0.81	0.81	200



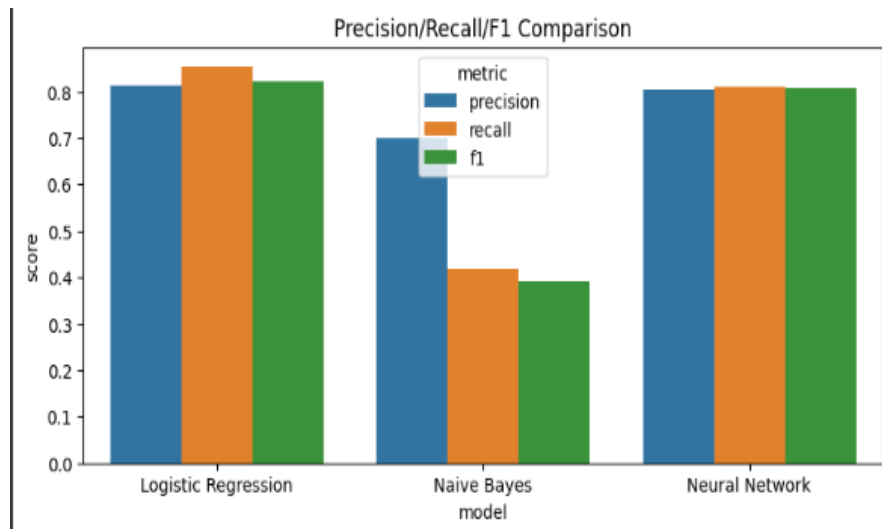
Accuracy comparison:



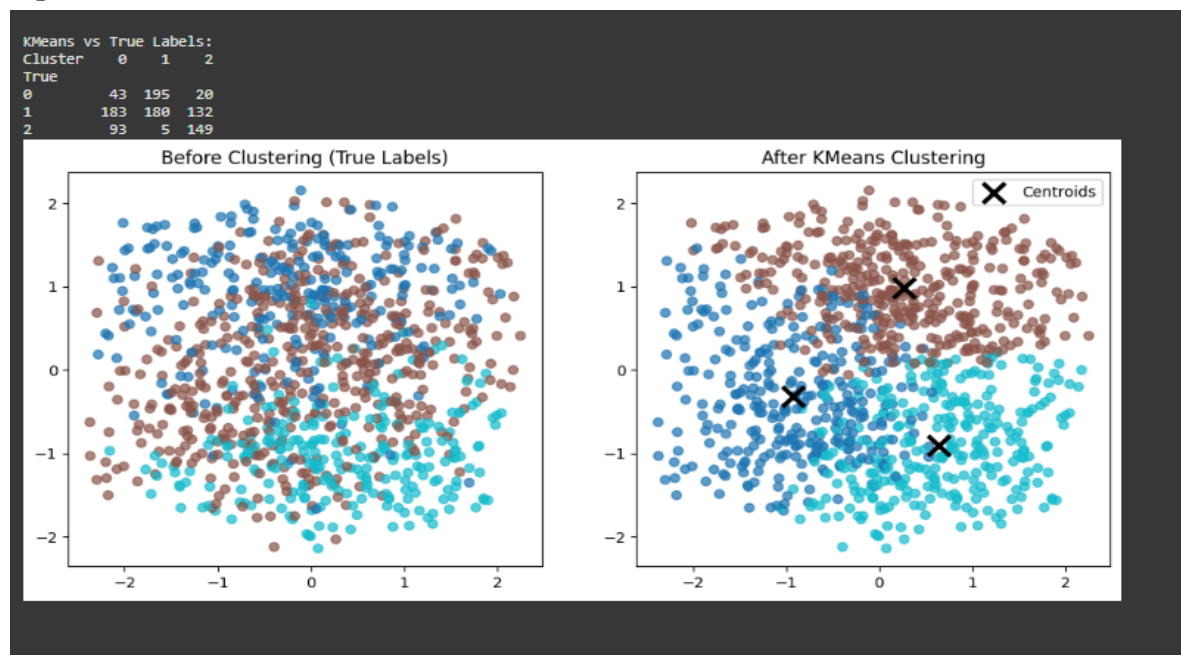
ROC-AUC Analysis: Logistic Regression achieved the highest AUC (0.94), followed closely by the Neural Network (0.93), while Naive Bayes performed the weakest with an AUC of 0.72.



Precision, Recall, and F1-Score Comparison: Logistic Regression achieved the best overall balance with precision 0.81, recall 0.85, and F1-score 0.82. The Neural Network performed closely with similar values, while Naive Bayes lagged behind with much lower recall (0.41) and F1-score (0.39).



KMeans Clustering results: KMeans formed three clusters that partially aligned with the true delivery speed labels. While some overlap exists between classes, the clustering still revealed meaningful groupings, with centroids capturing the main distributions. This shows that unsupervised learning can identify patterns, but it is less accurate than supervised models.



Conclusion:

This project aimed to predict food delivery times using factors such as distance, preparation time, courier experience, weather, and traffic with different machine learning models. After preprocessing the dataset through handling missing values, encoding categorical features, and applying scaling, three supervised models Logistic Regression, Naive Bayes, and Neural Network along with KMeans clustering were evaluated.

Logistic Regression emerged as the most reliable model, offering the best balance across accuracy, precision, recall, and F1-score. The Neural Network also performed strongly but required more computational effort, while Naive Bayes struggled due to its independence assumptions. KMeans revealed some group patterns but was less effective than supervised methods.

These results highlight the importance of careful model selection in predictive analytics. Logistic Regression's robustness makes it well-suited for this task, while Neural Networks remain a strong alternative. Overall, the project shows how machine learning can support food delivery services in improving efficiency and customer satisfaction.